

LETTER • OPEN ACCESS

Do we accurately measure what we consume?

To cite this article: Benjamin P Goldstein *et al* 2024 *Environ. Res. Lett.* **19** 084006

View the [article online](#) for updates and enhancements.

You may also like

- [Greenhouse gas emissions and energy use associated with production of individual self-selected US diets](#)
Martin C Heller, Amelia Willits-Smith, Robert Meyer et al.
- [An assessment of individual foodprints attributed to diets and food waste in the United States](#)
Catherine I Birney, Katy F Franklin, F Todd Davidson et al.
- [Differentiated responsibilities of US citizens in the country's sustainable dietary transition](#)
Oliver Taherzadeh and Keiichiro Kanemoto

ENVIRONMENTAL RESEARCH
LETTERS

LETTER

Do we accurately measure what we consume?

OPEN ACCESS

RECEIVED
8 April 2024REVISED
14 June 2024ACCEPTED FOR PUBLICATION
25 June 2024PUBLISHED
9 July 2024

Original content from
this work may be used
under the terms of the
[Creative Commons
Attribution 4.0 licence](#).

Any further distribution
of this work must
maintain attribution to
the author(s) and the title
of the work, journal
citation and DOI.

Benjamin P Goldstein^{1,*} , Dimitrios Gounaridis¹ , Joshua P Newell¹ , Rylie Pelton²
and Jennifer Schmitt² ¹ School for Environment and Sustainability, University of Michigan, Ann Arbor, MI, United States of America² Institute on the Environment, University of Minnesota, St. Paul, MN, United States of America

* Author to whom any correspondence should be addressed.

E-mail: benjgo@umich.edu**Keywords:** sustainable consumption, meat, environmental footprints, energy, diets, transportSupplementary material for this article is available [online](#)

Abstract

Understanding how consumption patterns affect the environment and shape well-being hinges on the rationale that the data collected on what is consumed, who consumes it, and where it is consumed are indeed accurate. To identify these consumption patterns and recommend corresponding policies, researchers and policy makers often rely on national surveys. Studies have explored the accuracy of individual surveys and the level of agreement across surveys of the same type (e.g. household expenditures), but no studies have compared representative national surveys measuring consumption in different ways. This study compares household consumption measured as expenditures and as material consumption (i.e. physical units) to assess how well we currently measure what we consume. We use multiple rigorous, national surveys to estimate meat consumption, household energy use, and private automobile use in the United States, with consumption profiles parsed by affluence, race/ethnicity, and education. Our results indicate that commonly used surveys may not accurately track important aspects of household consumption. For meat consumption, which included 30 consumption profiles detailing the consumption patterns across different demographic characteristics and meat types (e.g. kilograms beef consumed/capita for Caucasians), there is considerable disagreement between data sources for 20 profiles. By contrast, national surveys accurately measure household energy and transport (disagreement for four profiles). Our findings indicate that national surveys more accurately measure consistently tracked, standardized consumables like electricity than irregularly tracked, variable goods such as food. These results cast doubt on studies that use national surveys to draw conclusions about the how the environmental impacts of food, and, potentially, other goods (e.g. manufactured goods) vary across demographic groups. Overcoming this challenge will necessitate new surveys, updating legacy databases, and harnessing breakthroughs in data science.

1. Introduction

Reducing society's environmental footprint requires significant changes to household consumption patterns [1, 2]. Meat consumption, for instance, is projected to grow globally by almost 75% by 2050 and has a broad influence on human and planetary health [3, 4]. Developing strategies to address the impacts of human activities, from eating to mobility, hinges on accurate data of household and individual consumption patterns.

Consumption data come primarily from national surveys. In the United States (U.S.), examples include the Consumer Expenditure Survey (CEX) [5], the National Health and Nutrition Examination Survey (NHANES) for diets [6], the National Household Travel Survey for transport (NHTS) [7], Residential Energy Consumption Survey (RECS) for household energy [8], and the American Time Use Survey [66].

In place of large-scale national efforts, some researchers have crafted surveys to assess household consumption [9]. However, these are not broadly

representative and only provide a snapshot since they typically occur once. Utility data, as explored by Pincetl and Newell [10], offer insights into consumption, but can be costly and challenging to obtain. Alternative estimation techniques utilizing statistical models, engineering, or physiological approaches are often too complex to be scalable or rely on national survey data themselves [e.g. 11]. Consequently, national surveys continue to be the primary data source for analyzing consumption patterns and effects.

Towards this end, researchers use these surveys to create consumption profiles of the amounts and types of goods, energy, and services consumed by respondents, demonstrating that these profiles can vary significantly by affluence, ethnicity, age, and other traits [5, 12–15]. Geodemographic expenditure profiles derived from national household surveys and sold by companies also exist for select countries [16]. Researchers use these consumption profiles to identify the social, economic, cultural, geographic, and environmental determinants of consumption as well as to calculate the environmental impact of different groups [17, 18]. Researchers have long been aware of the issues of using expenditure data to estimate material consumption [19, 20]. But since these errors do not affect the accuracy of models of per capita impacts *between* countries, most critiques have focused on the models themselves rather than data on consumption *per se* [21]. However, researchers increasingly study the impacts of consumption by demographic groups *within* countries using national surveys (or derivatives) [22–26].

But what if national surveys do not accurately capture differences between demographic groups? This would hamper efforts to understand how and why the impacts of consumption vary between population groups. There has been surprisingly little scholarship asking this question. Kilian *et al* [16] found moderate similarity in carbon footprints estimated using three sources of survey data in the United Kingdom, but this analysis only compared *expenditure* data. Other national surveys, such as dietary surveys, record material consumption in physical units. Analysis of these surveys suggest that they underestimate certain aspects of consumption (e.g. red meat) [27].

Although tracking consumption is complex and bound to have uncertainties, consumption profiles should generally agree across both methods if we are to trust the findings of studies that use these data. Aside from a single study that found divergence between food expenditures and caloric intake amongst retirees [28], there have been few comparisons between national surveys in the literature. But what if meaningful discrepancies exist between consumption patterns recorded as expenditures and those measured in physical units? Such differences

would complicate drawing reliable conclusions about the environmental footprints of different demographic groups. There is a critical need for analysis of national survey data, especially as these studies continue to inform policy and influence popular discourse about consumption and its sustainability [29, 30].

Here, we investigate the quality of national consumption survey data. By consumption, we refer to final consumption by households and individuals in their daily activities, not intermediate consumption by businesses. We focus on how well these surveys measure material consumption (e.g. watts of energy, kilograms of food), as this is what drives environmental change and determines if households are meeting basic needs.

We analyzed survey data for three consumption categories; food (focusing on meat), household energy, and transport (focusing on automobile use). We assess these categories because of their significant environmental impacts and the key differences in how they are measured and tracked. Household energy and automobile use are consumed as commodities (e.g. electricity or fuel blends) and are consistently calculated by electricity meters or fuel gauges. This contrasts markedly with meat (and food generally) which is consumed in a variety of forms (e.g. beef as steak vs. Bolognese) and contexts (e.g. at home vs. restaurant). In this study, we focused on the U.S. where per capita consumption and related environmental impacts are amongst the highest globally and where there are longstanding dedicated government surveys in each consumption category.

To assess data quality, we compared consumption estimates using two data forms. The first are expenditures taken from the CEX, which includes data on the U.S. dollars spent on each of the three investigated categories. The second are national surveys that measure material consumption in physical units for each category.

After developing the consumption profiles based on race/ethnicity, affluence, and level of education, we assessed the degree of agreement between consumption as measured by expenditures vis-a-vis consumption measured through physical units. General agreement between the expenditure and physical consumption was interpreted as adequate measurement in a given category.

Our results indicate that household energy and private automobile use are consistently measured, but meat consumption is not. New surveys that record physical consumption may more accurately capture meat and similar forms of consumption, as might applying data-driven approaches to novel data. Overcoming these shortcomings is critical to understanding household consumption and thus developing effective policies to reduce related environmental impacts.

2. Methods

We used four surveys to estimate consumption of meat, household energy, and transport. Note that our goal was to gauge agreement between the surveys and not to analyze consumption trends or drivers for different sub-populations, as this has been done extensively elsewhere [e.g. 11, 31, 32].

All four surveys were designed to be nationally representative by accounting for geographic variation and ensuring adequate sampling of minorities. Each survey employs multiple layers of quality control and undergoes data validation. For instance, participants are given tools and training to measure food portions in NHANES and extreme values are investigated [33]. Interviewers are also retrained annually and data entry is monitored for accuracy and completeness [34]. Other measures, such as daily sampling, reduce systemic bias introduced by day-to-day variation in consumption patterns [35]. Full details about the quality control and data validation methods for each survey are available on the NHANES [6], NHTS [7], RECS [8], and CEX [5] websites.

In each survey, data were collected throughout the entire year for multiple years to capture seasonal and short-term shifts in consumption patterns. We selected these surveys for their advanced sampling methods, for their focus on consumption activities that strongly influence environmental impacts of individuals, and for their widespread use by domestic and international researchers to investigate the sustainability of consumption. Moreover, a primary purpose of these surveys is to parse out differences in consumption and expenditure patterns between sub-populations. Below we provide more details about each survey and how we utilized the data. Additional details on the surveys can be found in table S1.

2.1. Estimating meat

We estimated meat consumption using two nationally curated surveys; NHANES covering 51 623 individuals between 1999 and 2008, and the CEX for the years 2009 and 2010 covering 71 893 individuals (~28 000 households). Survey years were chosen to be as close as possible to avoid the influence of longer-term dietary trends (e.g. declining beef consumption since the 1970s) [31, 36] and facilitate comparisons between datasets. Using these survey years also allowed us to link NHANES to recipe databases to estimate meat consumption (see below). We estimated per capita beef, chicken and pork consumption using the prescribed protocols for each survey and additionally compared to estimates found in the literature.

NHANES is recorded as the mass of meals consumed. Meals are recorded as standardized U.S. Department of Agriculture (USDA) food codes. We linked NHANES responses to the USDA Food Intake

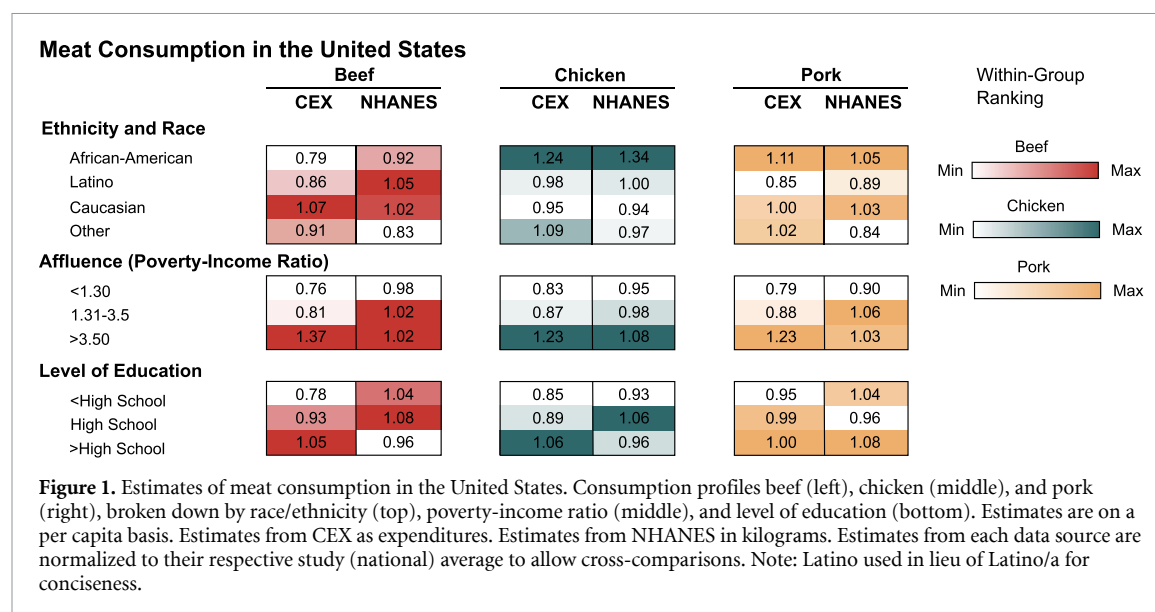
Converted to Retail Commodity database (FICRCD) [37] to disaggregate meals to individual meats. The FICRCD is specifically designed for NHANES and gives the grams of beef, chicken and pork per 100 grams of USDA food code which we used to estimate the mass of per capita consumption of each meat. To align with the FICRCD, we selected NHANES cycles from 1999 to 2008. Despite their age, these data were collected using modern survey techniques (computer-assisted personal interviews) and largely follow current NHANES sampling protocols.

CEX estimates of meat expenditures by demographic groups were developed using prescribed survey protocols. We include four races/ethnic groups, three levels of affluence (taken as ratio of income to federal poverty level), and three levels of education, producing 30 consumption profiles (three meats and 10 demographic groups).

Expenditures on beef, chicken, and pork consumed at home were identified using six-digit Universal Classification Codes which provide descriptions of thousands of specific expenditures. We estimated total consumption of each meat using the ratio of at home expenditures to total expenditures on food (e.g. total chicken = kg of chicken in home $\times$ total food expenditures/food expenditures at home). This method erases price differentials between in- and out-of-home consumption. Unlike NHANES, the CEX does not provide the poverty-income ratio of respondents. Poverty-income ratios were taken as the ratio of the household income divided by the federal poverty threshold for that household size in the respondent's state. Raw data and model outputs are available on Open Science Framework: <https://shorturl.at/hxM56>.

2.2. Estimating greenhouse gas emissions from meat

We used two common methods for calculating consumption-side environmental footprints to clarify how uncertainty in consumption data might influence study findings. For NHANES, which provides consumption in mass, we used process-based estimates of median greenhouse gas (GHG) intensities for U.S. meat production in kilograms carbon dioxide equivalents per kilogram meat (kg CO₂e/kg) for beef (26.4 kg CO₂e/kg), chicken (5.0 kg CO₂e/kg), and pork (6.9 kg CO₂e/kg) [38]. Mass consumed of each meat was multiplied by its GHG intensity and summed to get total GHGs from meat consumption per capita for each demographic group. For CEX estimates, we combined expenditures with the U.S. Environmentally-Extended Input-Output Model v.2.0 [39]. These were normalized to average GHGs from U.S. meat consumption for each method to facilitate comparisons across methods and data. Both the mass-based and input-output approaches are widely used in the literature [40–42].



2.3. Estimating household energy use

We estimated household natural gas, fuel oil, and electricity use with the 2015 RECS (5 686 households) [8] and household energy expenditures with the 2015 CEX (26 697 households interviewed, 11 745 households surveyed) [5]. RECS estimates of annual energy consumption at the household level were calculated using the Survey package in R to account for the complex survey design of the RECS. CEX estimates of annual household energy expenditures were taken directly from the published CEX tables. The consumption profiles mirror those developed for meat, except for affluence, where we group households into low, medium, and high, because the RECS lacks the geographic specificity needed to calculate poverty-income ratio.

2.4. Estimating automobile use

We estimated private vehicle use with the 2017 NHTS (129 696 households) and gasoline expenditures with the 2017 CEX (24 192 households interviewed, 11 659 households surveyed). NHTS provides self-reported vehicle miles for drivers. To estimate household level vehicle miles per household in a population group, for each group we multiplied vehicle-miles per driver by the number of drivers and then divided by number of households. CEX estimates of annual household gasoline expenditures were taken directly from the published CEX tables. The NHTS does not provide information on poverty-income ratio of respondents. For affluence, using household income, we grouped households into low, medium, and high income (different cutoffs used for NHTS and RECS due to differences in how data are reported). It was not possible to estimate vehicle-miles based on level of education because NHTS does not provide number of households for this group.

3. Results

3.1. Meat Consumption in the United States

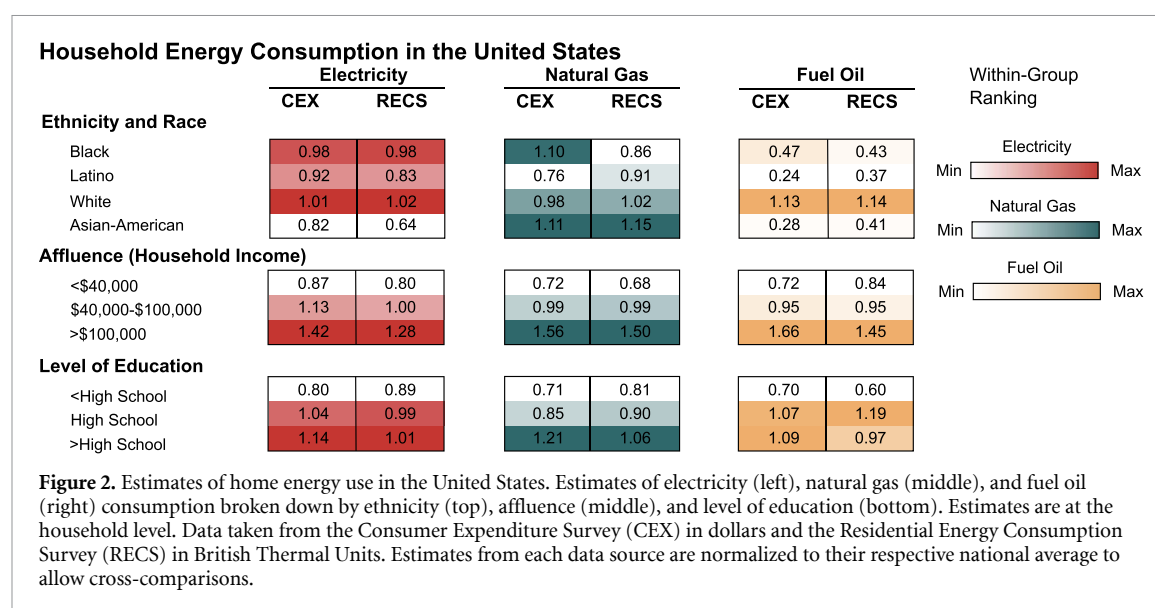
We find general disagreement between the CEX and NHANES estimates of meat consumption. This aligns with earlier work that found CEX disagreed with NHANES by underestimating caloric intake by older adults [28]. Figure 1 shows the range of estimates of beef, chicken, and pork consumption by each consumption profile (each dataset's estimate shown as a ratio of its respective national average), indicating whether the demographic variable is above or below average consumption. Results differ by up to 35% across data sources and rankings of demographic groups are inconsistent. Twelve consumption profiles show a demographic group to be consuming both above and below average amounts of meats, and rankings disagree for 20 groups, depending on the survey data used.

3.1.1. Ethnicity and race

Estimates for the three types of meat range on average 12% for a given demographic group across surveys (figure 1). Beef consumption by the Latinos has the largest range (19%). The smallest range is for chicken consumption by Caucasian individuals (1%). Results for White individuals shows the least variation (ranges between: 1% and 5% across the three types of meat). Estimates are both above and below the national average for beef consumption by Latinos, chicken and pork consumption by Others.

3.1.2. Affluence

Estimates for the three types of meat range on average 18% for a given demographic group across surveys (figure 1). Beef consumption by the high-affluence group has the largest range (35%). The smallest



range is for chicken consumption by medium-affluence individuals and pork consumption by low-affluence individuals (11%). Results for medium-affluence shows the least variation (ranges between: 11% and 22% across the three types of meat). Estimates are both above and below the national average for beef and pork consumption by middle-income individuals.

3.1.3. Level of education

Estimates for the three types of meat range on average 12% for a given demographic group (figure 1). Chicken consumption by individuals without a high-school diploma has the largest range (26%). Pork consumption by individuals with a high-school education has the smallest range (3%). Results for individuals with some post-secondary education show the least variation within the demographic group and across all demographic groups investigated (ranges between 8% and 10%).

Beyond large variation, seven of nine estimates are both above and below the national average and are thus not even directionally consistent. Exceptions are chicken consumed by low-education households (below average) and pork consumption by households with high-school education (below average).

3.2. Household energy use in the United States

Figure 2 shows estimates of electricity, natural gas, and fuel oil consumption. For household energy use, we find good agreement between the datasets. The consumption profiles span smaller ranges than those of meat consumption (Average range: 9%. Max range: 24%). Importantly, rankings are much more consistent across the two data sources for energy use compared to meat consumption, with agreement for 25 of the 30 consumption profiles. There are only four

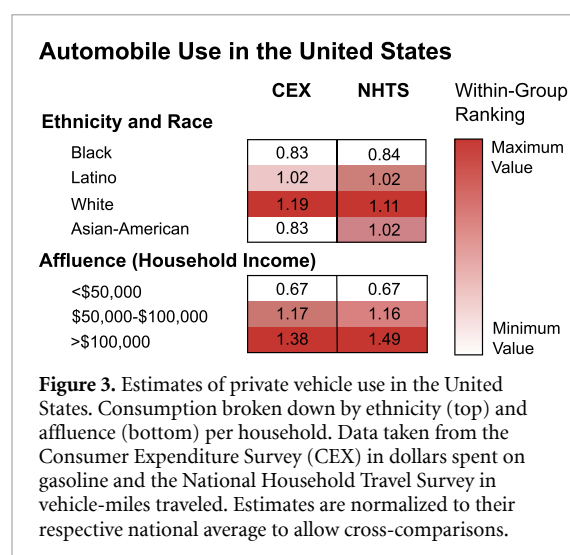
instances where a given sub-population's consumption goes from above to below national average or vice versa when comparing surveys. This shows partial agreement with Kilian *et al* [16] who found that expenditures poorly tracked natural gas consumption but were a suitable proxy for electricity when comparing different expenditure datasets in the United Kingdom.

3.2.1. Ethnicity and race

Consumption estimates of natural gas, fuel oil, and electricity by demographic groups range on average 8% (figure 2). Natural gas consumption by African-Americans has the largest range (24%). We found no differences between data sources for electricity consumption by African-Americans and for fuel oil consumption by Caucasians. Results for Caucasians show the least variation (ranges between: 1% and 3%) and cluster around the national average. We found inconsistencies for natural gas by ethnicity, where African-Americans drop from second highest in the CEX to the lowest consuming group in the RECS, and Latinos go from lowest in the CEX to second highest in the RECS. Natural gas consumption by African-Americans and by Caucasians are the only estimates that are both above and below the national average.

3.2.2. Affluence

Consumption profiles vary on average 9% between data sources. Fuel oil use by high-income households had the largest variation between sources (21%). Natural gas and fuel oil consumption by middle-income households had the lowest variation (0%). Rankings are consistent across all consumption profiles. There are no instances where consumption by a demographic group goes from above to below the national average.



3.2.3. Level of education

Consumption estimates range on average 10% for a given demographic group and energy source (figure 3). Natural gas consumption by high-education individuals has the largest range (16%). Electricity and natural gas consumption by African-Americans have the smallest ranges (5%). Rankings are consistent except for fuel oil, where households with post-secondary and high-school degrees both rank first or second depending on data source. Electricity consumption by households with high-school degrees and fuel oil consumption by households with post-secondary education are estimated to be both above and below the national average.

3.3. Automobile use

Figure 3 shows estimates of private vehicle use, measured as vehicle-miles traveled (VMT) per household, for demographic groups based on race/ethnicity and affluence. For automobile use, we find strong agreement between the datasets. The consumption profiles span small ranges, with an average range of 6%. Asian-American households (category unavailable for our food and household energy comparisons) have the highest disagreement between data sources (19%). Estimated VMT is identical between data sources for Latino and low-income households. Rankings are consistent across the two data sources, with agreement across all consumption profiles, except African-American and Asian-American households, which have either lowest or second lowest VMT depending on the survey. Asian-American households are the only consumption profile to go from above to below national average depending on the data source.

3.4. Explaining agreement between datasets

Disagreements, primarily in ranking, between consumption-profiles suggests mismeasurement. For meat consumption, differences in survey methods

are likely important but should not be so influential. Both NHANES and CEX are nationally representative, use diaries and computer-assisted interviews to collect data, and sample throughout the year to account for seasonal variation. One key difference is that NHANES details the types of foods eaten away from home while the CEX does not. Nonetheless, there are other factors at play.

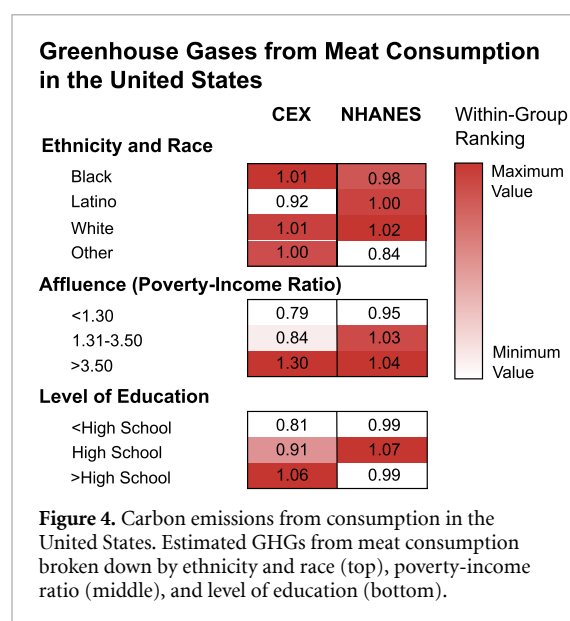
Like all self-reported data, NHANES and CEX are susceptible to underestimating consumption of certain foods, including meat [27]. The CEX faces additional challenges. Differential pricing of meat means that the same expenditure can purchase vastly different quantities of meat (e.g. ground beef vs. Wagyu or at home vs. restaurant). This can overstate correlations between income and meat consumption, since wealthy households that spend more on meat may be purchasing more expensive cuts, not consuming more. Although meat intake and income are strongly correlated at low incomes, this relationship becomes weaker in high-income countries like the U.S [3, 43]. Our analysis of NHANES data suggests the opposite trend or an attenuated relationship between affluence and the consumption of beef [44] and pork [45].

Though based on different data, consumption profiles of household energy use from the RECS and CEX largely agree. The RECS uses utility bills to record annual energy use. The CEX also uses utility bills, but it captures energy expenditures. Consistent pricing and unified record keeping through regular utility bills make expenditures an adequate proxy for consumables like household energy. Inconsistencies between RECS and CEX for natural gas and fuel oil estimates suggest that these data are problematic. Differential pricing across a large country like the U.S. could be one explanation as could seasonal variation (e.g. demand for heating during winter when prices are higher than annual average).

The CEX also measures automobile use adequately. The NHTS uses trip logs and odometer data to estimate VMT by private vehicle. For the CEX, gasoline receipts are recorded in a household expenditure diary. Given the clear link between VMT and gasoline consumption, and the standard pricing of gasoline, alignment between the CEX and NHTS is unsurprising. Regional gasoline price variations should be minimal given that oil is a global commodity, although seasonality could be important for the same reasons as home energy use.

4. Discussion

Robust data are essential to tackling unsustainable consumption. Meat consumption is challenging to measure because it is often one component in a meal with many ingredients. Moreover, it is sporadically recorded (e.g. receipts of various origin and quantities that are not easily aggregated), and typically in



monetary units that are, at best, a shaky proxy for physical consumption across contexts due to variable pricing (e.g. grocery store vs. fast food outlet vs. Michelin-starred restaurant) and aggregation (e.g. prepared meals with multiple ingredients).

Food, durable goods, and services are other areas of consumption where branding, design, or other intangible properties blur the link between expenditures and physical consumption. For instance, haute-couture or designer clothes cost much more than fast-fashion reproductions. Similarly, one can find large price gaps between computers and phones with nearly identical components and functionality marketed under different brands. In the U.S., and many other countries, expenditure surveys are the only way that most of this consumption is tracked. As our results for meat reveal, it is questionable whether this adequately captures important aspects of material consumption.

4.1. Implications of inadequate measurement

Poor data complicates interpreting studies that use national survey data. Most environmental footprint methods rely heavily on expenditures as a proxy for consumption. CEX data are suitable when studying national averages or select aspects of consumption, but researchers increasingly use these models to explore how income [41, 46–48], race/ethnicity [25], sex [49], and other demographic characteristics [42, 50] relate to environmental impacts for broad categories of consumption. Others have used CEX to show that consumption by certain racial/ethnic groups cause disproportionate air pollution burdens for other groups [51]. Likewise, studies using physical data, such as NHANES, to understand impacts across demographic groups may also be basing conclusions on inconsistent data [40].

To demonstrate this, figure 4 compares carbon footprints of U.S. meat consumption profiles. Again,

we see gaps between CEX and NHANES that would lead to different conclusions depending on the survey used. The CEX suggests significant variation across economic groups. NHANES shows only slight differences between consumption profiles. In the CEX, education and GHGs from meat rise in tandem, contradicting NHANES and the numerous dietary surveys that suggest that the consumption of GHG-intensive red meat is lower in more educated households. Based on carbon footprints using CEX data, one might suggest a tax to shift consumption when educational programs might be more effective and equitable [52]. Meat is but one area of consumption. If similar data challenges exist for other consumptive activities, then we might be inaccurately characterizing nontrivial components of the environmental impacts of households and individuals [16].

4.2. More accurately measuring consumption

Our results suggest that national surveys accurately measure keys aspects of consumption, such as transport and energy, but are less reliable for other areas. This does not invalidate consumption-side footprint studies, as the input data still accurately capture key components of household consumption. Moreover, surveys serve different needs and may be adequate for their intended purposes (e.g. tracking nutrient intake with NHANES or tracking inflation with the CEX). But given the urgency to reduce environmental impacts across all sectors, more accurate data are needed for many areas of consumption.

Measuring consumption better requires addressing the multiple challenges of complex mixes of materials, inconsistent measurement, and scattered record keeping. For food, dietary surveys currently provide the best basis for measuring consumption. Dietary surveys record consumption in physical units, collect data from large sample populations, and use standardized data collection protocols. Nonetheless, NHANES suffers from memory recall errors and respondent bias [53]. Even if survey errors are eliminated, a remaining challenge lies in identifying the types and amounts of ingredients.

Prior to 2008, NHANES could be linked to the FICRCD, which converts meals recorded in NHANES to 65 foods (e.g. beef, spinach, etc) for nearly 10 000 meals including food waste during preparation [37]. This was superseded by databases better aligned with NHANES remit of tracking nutrition, including the Food Pattern Equivalents Database [37] which converts meals to broad nutritional categories like ‘red meat’ and ‘dark-green vegetables.’ While fit for purpose, the degree of aggregation hinders efforts to curtail specific ingredients for reasons of health or the environment. Beef, for example, is grouped with pork as ‘red meat,’ but is substantially more environmentally intensive to produce [54]. The USDA Food and Nutrient Database for Dietary Studies [55], which provides nutrient content of meals, does identify

specific meats in meals, but is cumbersome and excludes food losses.

Revitalizing the older database would be practical for tracking U.S. food consumption. Similar efforts to link nutritional surveys to recipe databases are needed in other countries. Ideally these methods should be standardized internationally to facilitate cross-comparisons [56]. National health authorities that administer these surveys, like the U.S. Centers for Disease Control, could augment existing surveys or develop new ones to better track consumption of criteria foods (akin to criteria air pollutants) that have significant health and environmental impacts. Academics have developed one-off surveys that do this [57].

To adequately measure electronics, clothes, and durable goods, new data gathering initiatives are needed. Prior to NHANES, the U.S. dietary survey was called 'What We Eat in America.' A 'What We Consume in America' survey could track the consumption of durable goods by different sub-populations and build off the CEX. The U.S. Bureau of Labor Statistics, having recognized that expenditures do not always reflect consumption of resources [58], is developing a new consumption metric to complement the CEX [59]. An additional initiative to track physical consumption could augment this.

For example, CEX diaries and interviews could be updated to record more specifics on purchased products (e.g. brand or model). This data could be linked to a database containing the masses of consumer goods and 'recipes' of the dozens of materials they contain [60, 61]. Although akin to existing data used in life cycle assessments (e.g., Ecoinvent), the accuracy and comprehensiveness of our proposed dataset would require industry collaboration. For instance, online retailers (e.g. Amazon, Alibaba) could provide data on the weights and compositions of the millions of specific, branded products they sell, as could manufacturers. Funding agencies could catalyze required data collection and integration. Large-language models could automate this laborious process.

Other methods can strengthen or supplement surveys. Academics have developed accurate online surveys of consumption with free websites [62, 63], though these lack the breadth and representativeness of national efforts. Governments have also experimented with web- and computer-based dietary surveys to reduce respondent error and improve accuracy [64]. Artificial intelligence can accurately identify amounts and types of materials in photos (e.g. ingredients in meals) [65]. This technology could semi-automate and standardize record keeping and address the chronic underreporting of meat and other ingredients.

New forms of expenditure data may also prove useful. Scanner data can track purchases of thousands of products at sub-national scales and at varying

timescales (www.nielsen.com). But these data do not always provide clear linkages to demographic groups. Credit card 'spend' data provide information on spending patterns by demographic groups (www.secondmeasure.com), but are expensive, prone to sampling bias, and must be supplemented with data on cash and debit purchases. Ultimately, adequately measuring consumption will likely necessitate combining various datasets measured as expenditures and physical units [16].

Now is the time for funders of research and researchers to work collectively to better measure what we consume. Wealthy countries already consume resources at unsustainable levels and are beset by high rates of lifestyle related diseases that stem from unhealthy choices surrounding food, mobility, and leisure. With consumption patterns of billions of aspirational consumers in low- and middle-income countries changing, it is critical to promote sustainable choices before unsustainable habits become locked-in.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://doi.org/10.17605/OSF.IO/E6NMQ>.

Acknowledgments

Thanks to the reviewers for their time and insightful comments.

Funding information

We gratefully acknowledge the financial support of this work by the National Science Foundation through the Environmental Sustainability Program (Award 1805085).

ORCID iDs

Benjamin P Goldstein  <https://orcid.org/0000-0003-0055-1323>

Dimitrios Gounaridis  <https://orcid.org/0000-0003-3145-7284>

Joshua P Newell  <https://orcid.org/0000-0002-1440-8715>

Rylie Pelton  <https://orcid.org/0000-0002-6015-2263>

Jennifer Schmitt  <https://orcid.org/0000-0002-5427-5208>

References

- [1] Creutzig F *et al* 2018 Towards demand-side solutions for mitigating climate change *Nat. Clim. Change* **8** 268–71
- [2] Wiedmann T, Lenzen M, Keyßer L T and Steinberger J K 2020 Scientists' warning on affluence *Nat. Commun.* **11** 1–10
- [3] Tilman D and Clark M 2014 Global diets link environmental sustainability and human health *Nature* **515** 518–22

- [4] Godfray H C J, Aveyard P, Garnett T, Hall J W, Key T J, Lorimer J, Pierrehumbert R T, Scarborough P, Springmann M and Jebb S A 2018 Meat consumption, health, and the environment *Science* **361** eaam5324
- [5] U. S. Bureau of Labor Statistics 2024 *Consumer Expenditure Surveys - Tables* (available at: <https://www.bls.gov/cex/tables.htm>)
- [6] Centers for Disease Control and Prevention 2020 *National Health and Nutrition Examination Survey* (available at: www.cdc.gov/nchs/nhanes/index.htm)
- [7] U. S. Department of Transportation-Federal Highway Administration 2020 *National Household Travel Survey*
- [8] U. S. Energy Information Administration 2017 Residential energy consumption survey (RECS) (available at: www.eia.gov/consumption/residential/index.php)
- [9] Pandey B, Reba M, Joshi P K and Seto K C 2020 Urbanization and food consumption in India *Sci. Rep.* **10** 17241
- [10] Pincetl S and Newell J P 2017 Why data for a political-industrial ecology of cities? *Geoforum* **85** 381–91
- [11] Goldstein B, Gounaridis D and Newell J P 2020 The carbon footprint of household energy use in the United States *Proc. Natl Acad. Sci.* **117** 19122–30
- [12] Panzone L, Hilton D, Sale L and Cohen D 2016 Socio-demographics, implicit attitudes, explicit attitudes, and sustainable consumption in supermarket shopping *J. Econ. Psychol.* **55** 77–95
- [13] Huebner G, Shipworth D, Hamilton I, Chalabi Z and Oreszczyn T 2016 Understanding electricity consumption: a comparative contribution of building factors, socio-demographics, appliances, behaviours and attitudes *Appl. Energy* **177** 692–702
- [14] Clonan A, Roberts K E and Holdsworth M 2018 Socioeconomic and demographic drivers of red and processed meat consumption: implications for health and environmental sustainability *Proc. Nutr. Soc.* **75** 367–73
- [15] Nowak J and Kochkova O 2011 Income, culture, and household consumption expenditure patterns in the European Union: convergence or divergence? *J. Int. Consum. Mark.* **23** 260–75
- [16] Kilian L, Owen A, Newing A and Ivanova D 2023 Microdata selection for estimating household consumption-based emissions *Econ. Syst. Res.* **35** 325–53
- [17] Young W, Hwang K, McDonald S and Oates C J 2010 Sustainable consumption: green consumer behaviour when purchasing products *Sustain. Develop.* **18** 20–31
- [18] Prothero A, Dobscha S, Freund J, Kilbourne W E, Luchs M G, Ozanne L K and Thøgersen J 2011 Sustainable consumption: opportunities for consumer research and public policy *J. Public Policy Mark.* **30** 31–38
- [19] Goldstein B P, Hauschild M Z, Fernández J E and Birkved M 2017 Contributions of local farming to urban sustainability in the northeast United States *Environ. Sci. Technol.* **51** 7340–9
- [20] Hertwich E G 2005 Life cycle approaches to sustainable consumption: a critical review *Environ. Sci. Technol.* **39** 4673–84
- [21] Rodrigues J F D, Moran D, Wood R and Behrens P 2018 Uncertainty of consumption-based carbon accounts *Environ. Sci. Technol.* **52** 7577–86
- [22] Chen G, Hadjikakou M, Wiedmann T and Shi L 2018 Global warming impact of suburbanization: the case of Sydney *J. Cleaner Prod.* **172** 287–301
- [23] Theine H, Humer S, Moser M and Schnetzer M 2022 Emissions inequality: disparities in income, expenditure, and the carbon footprint in Austria *Ecol. Econ.* **197** 107435
- [24] Mi Z, Zheng J, Meng J, Ou J, Hubacek K, Liu Z, Coffman D, Stern N, Liang S and Wei Y-M 2020 Economic development and converging household carbon footprints in China *Nat. Sustain.* **3** 529–37
- [25] Minx J, Baiocchi G, Wiedmann T, Barrett J, Creutzig F, Feng K, Förster M, Pichler P-P, Weisz H and Hubacek K 2013 Carbon footprints of cities and other human settlements in the UK *Environ. Res. Lett.* **8** 035039
- [26] Starr J, Nicolson C, Ash M, Markowitz E M and Moran D 2023 Income-based U.S. household carbon footprints (1990–2019) offer new insights on emissions inequality and climate finance *PLoS Clim.* **2** e0000190
- [27] Archer E, Hand G A and Blair S N 2013 Validity of U.S. Nutritional surveillance: national health and nutrition examination survey caloric energy intake data, 1971–2010 *PLoS One* **8** e76632
- [28] Aguiar M and Hurst E 2005 Consumption versus expenditure *J. Pol. Econ.* **113** 919–48
- [29] Lenzen M, Moran D, Kanemoto K and Geschke A 2013 Building Eora: a global multi-region input-output database at high country and sector resolution *Econ. Syst. Res.* **25** 20–49
- [30] Dawkins E 2019 Sustainable consumption for policymakers: measuring, learning and acting pp 13–16
- [31] Wells H and Buzby J 2008 Dietary assessment of major trends in U.S. food consumption, 1970–2005 (USDA) (available at: www.ers.usda.gov/media/210681/eib33_1_1.pdf)
- [32] McGuckin N and Fucci A (Westat Inc.) 2018 Summary of travel trends: 2017 national household travel survey *Report No.: FHWA-PL-18-019* (available at: <https://rosap.nhtl.bts.gov/view/dot/68751>) (Accessed 5 June 2024)
- [33] Centers for Disease Control and Prevention 2019 *NHANES—Measuring Guides* (available at: www.cdc.gov/nchs/nhanes/measuring_guides_dri/measuringguides.htm) (Accessed 5 June 2024)
- [34] Burcham S, Liu Y, Merianos A L and Mendy A 2023 Outliers in nutrient intake data for U.S. adults: national health and nutrition examination survey 2017–2018 *Epidemiol. Methods* **12** 20230018
- [35] Ahluwalia N, Dwyer J, Terry A, Moshfegh A and Johnson C 2016 Update on NHANES dietary data: focus on collection, release, analytical considerations, and uses to inform public policy *Adv. Nutr.* **7** 121–34
- [36] Daniel C R, Cross A J, Koebnick C and Sinha R 2011 Trends in meat consumption in the United States *Public Health Nutr.* **14** 575–83
- [37] U. S. Department of Agriculture 2019 Food intakes converted to retail commodities (available at: www.ars.usda.gov/northeast-area/beltsville-md/beltsville-human-nutrition-research-center/food-surveys-research-group/docs/ficrcd-overview/)
- [38] Heller M C and Keoleian G A 2015 Greenhouse gas emission estimates of U.S. dietary choices and food loss *J. Ind. Ecol.* **19** 391–401
- [39] Ingwersen W W, Li M, Young B, Vendries J and Birney C 2022 USEEIO v2.0, the US environmentally-extended input-output model v2.0 *Sci. Data* **9** 194
- [40] Bozeman J F, Springfield S and Theis T L 2020 Meeting EAT-lancet food consumption, nutritional, and environmental health standards: a U.S. case study across racial and ethnic subgroups *Environ. Justice* **13** 160–72
- [41] Feng K, Hubacek K and Song K 2020 Household carbon inequality in the US *J. Clean. Prod.* **278** 123994
- [42] Osorio P, Tobarra M Á and Tomás M 2024 Are there gender differences in household carbon footprints? Evidence from Spain *Ecol. Econ.* **219** 108130
- [43] Lin B H, Buzby J, Anekwe T and Bentley J 2016 *U.S. Food Commodity Consumption Broken down by Demographics, 1994–2008* (U.S. Department of Agriculture, Economic Research Service)
- [44] Davis C and Biting L 2005 Factors affecting US beef consumption (U.S. Department of Agriculture) (available at: http://webarchives.cdlib.org/wayback/public/UERS_ag_1/20110915064436/http://ers.usda.gov/publications/ldp/Oct05/ldpm13502/ldpm13502.pdf)
- [45] Davis C and Lin B H 2005 *Factors Affecting U.S. Pork Consumption* (U.S. Department of Agriculture) p 18
- [46] Song K, Qu S, Taiebat M, Liang S and Xu M 2019 Scale, distribution and variations of global greenhouse gas

- emissions driven by U.S. households *Environ. Int.* **133** 105137
- [47] Ivanova D and Wood R 2020 The unequal distribution of household carbon footprints in Europe and its link to sustainability *Glob. Sustain.* **3** 1–12
- [48] Wiedenhofer D, Guan D, Liu Z, Meng J, Zhang N and Wei Y M 2017 Unequal household carbon footprints in China *Nat. Clim. Change* **7** 75–80
- [49] Kanyama A C, Nässén J and Benders R 2021 Shifting expenditure on food, holidays, and furnishings could lower greenhouse gas emissions by almost 40% *J. Ind. Ecol.* **25** 1602–16
- [50] Heinonen J, Jalas M, Juntunen J K, Ala-Mantila S and Junnila S 2013 Situated lifestyles: i. How lifestyles change along with the level of urbanization and what the greenhouse gas implications are—a study of Finland *Environ. Res. Lett.* **8** 025003
- [51] Tessum C W et al 2019 Inequity in consumption of goods and services adds to racial–ethnic disparities in air pollution exposure *Proc. Natl Acad. Sci.* **116** 6001–6
- [52] Többen J, Pichler P P, Jaccard I S, Kratena K, Moran D and Zheng H 2023 Unequal carbon tax impacts on 38 million German households: assessing spatial and socio-economic hotspots *Environ. Res. Clim.* **2** 045006
- [53] Shim J S, Oh K and Kim H C 2014 Dietary assessment methods in epidemiologic studies *Epidemiol. Health* **36** 1–8
- [54] Eshel G, Shepon A, Makov T and Milo R 2014 Land, irrigation water, greenhouse gas, and reactive nitrogen burdens of meat, eggs, and dairy production in the United States *Proc. Natl Acad. Sci.* **111** 11996–2001
- [55] U. S. Department of Agriculture 2020 *Food and Nutrient Database for Dietary Studies* (available at: www.ars.usda.gov/northeast-area/beltsville-md-bhnrc/beltsville-human-nutrition-research-center/food-surveys-research-group/docs/fndds/)
- [56] Zezza A, Carletto C, Fiedler J L, Gennari P and Jolliffe D 2017 Food counts. Measuring food consumption and expenditures in household consumption and expenditure surveys (HCES). Introduction to the special issue *Food Policy* **72** 1–6
- [57] Wu Y, Wang X and Lu F 2012 The carbon footprint of food consumption in Beijing *Acta Ecol. Sin.* **32** 1570–7
- [58] Garner T 2021 Consumption as a resource measure *FCSM 2021 Research and Policy Conf.* (4 November 2021) (available at: https://www.bls.gov/cex/research_papers/pdf/garner-consumption-as-a-resource-measure.pdf)
- [59] Gerner T, Masumoto B, Schild J, Curtin S and Safir A 2023 Developing a consumption measure, with examples of use for poverty and inequality analysis: a new research product from BLS *Mon. Labor Rev.*
- [60] Dionisio K L, Phillips K, Price P S, Grulke C M, Williams A, Biryol D, Hong T and Isaacs K K 2018 Data Descriptor: the Chemical and Products Database, a resource for exposure-relevant data on chemicals in consumer products *Sci. Data* **5** 1–9
- [61] United States Department of Agriculture. USDA Food Composition Database 2020 *FoodData Central* (available at: www.nal.usda.gov/usda-food-composition-database)
- [62] Kalbar P P, Birkved M, Kabins S and Nygaard S E 2016 Personal metabolism (PM) coupled with life cycle assessment (LCA) model: Danish case study *Environ. Int.* **91** 168–79
- [63] Xue L, Liu X, Lu S, Cheng G, Hu Y, Liu J, Dou Z, Cheng S and Liu G 2021 China's food loss and waste embodies increasing environmental impacts *Nat. Food* **2** 519–28
- [64] Timon C M, Barg R V D, Blain R J, Kehoe L, Evans K, Walton J, Flynn A and Gibney E R 2016 A review of the design and validation of web- and computer-based 24-h dietary recall tools *Nutr. Res. Rev.* **29** 268–80
- [65] Yunus R, Arif O, Afzal H, Amjad M F, Abbas H, Bokhari H N, Haider S T, Zafar N and Nawaz R 2019 A framework to estimate the nutritional value of food in real time using deep learning techniques *IEEE Access* **7** 2643–52
- [66] U.S. Bureau of Labor Statistics 2024 *American Time Use Survey* (available at: <https://www.bls.gov/tus/>)